

STA 2101: Statistics & Probability

Lecture 06: Data Description (Ch 03)

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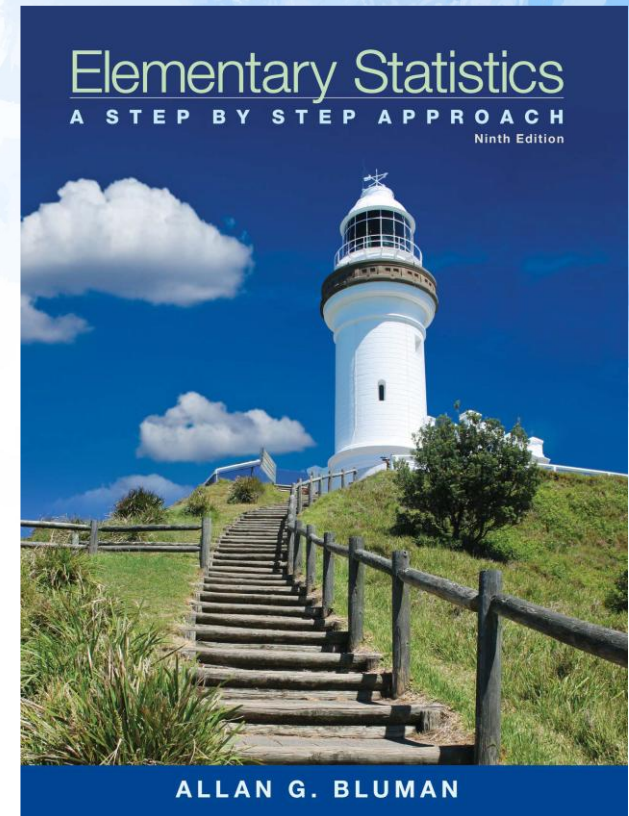
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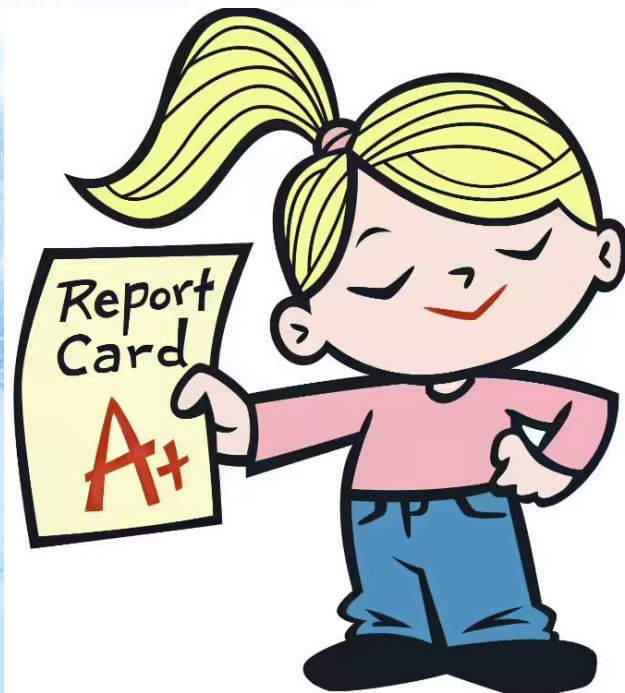
Fall 2025

Reference Book

- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

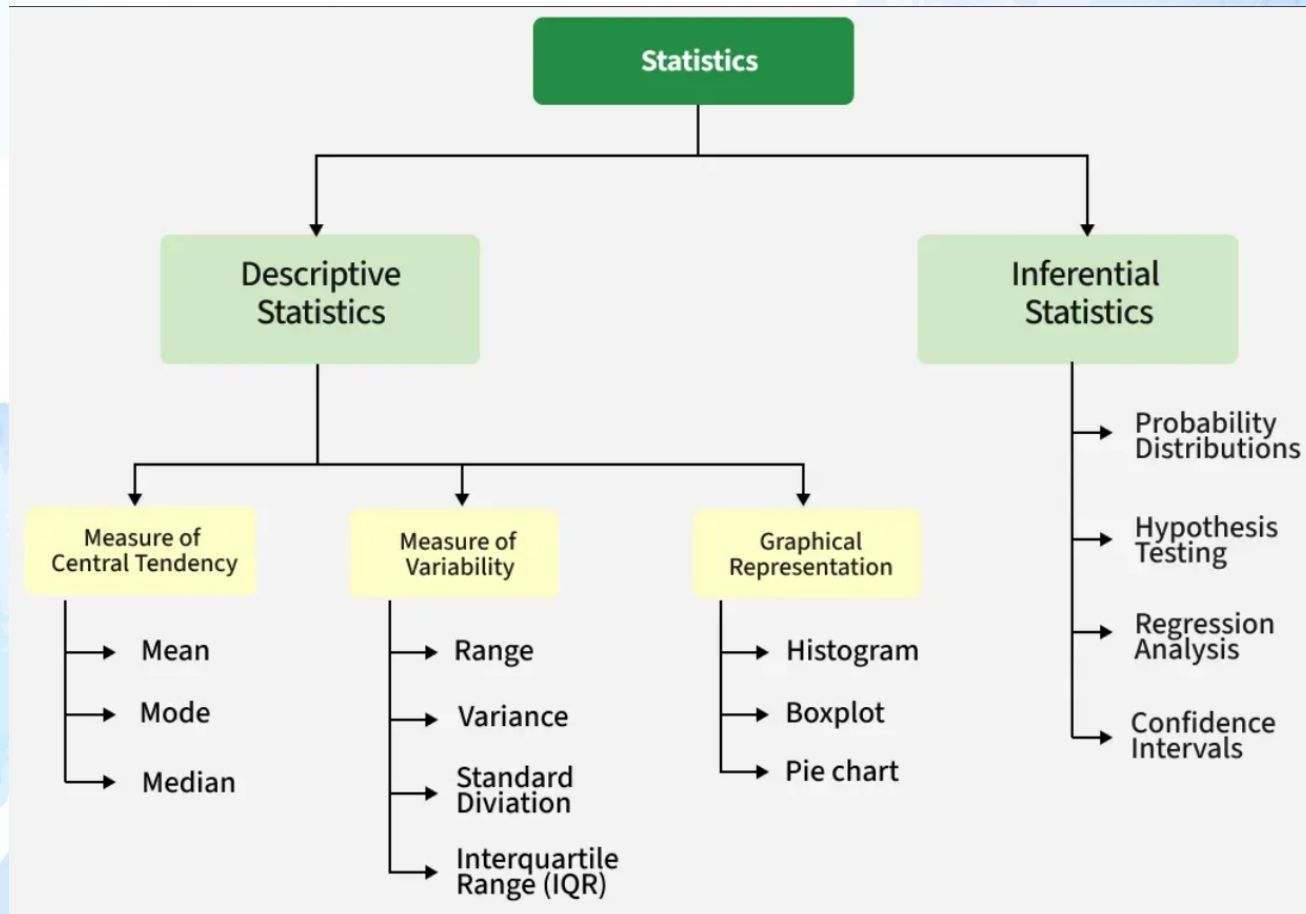


Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today's Topics

- Measures of Central Tendency
- Mean
- Median

Let's Rewind



The Mean

The **mean** is the sum of the values, divided by the total number of values.

The **sample mean**, denoted by $\bar{X}$ (pronounced “X bar”), is calculated by using sample data. The sample mean is a statistic.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} = \frac{\Sigma X}{n}$$

where n represents the total number of values in the sample.

The **population mean**, denoted by μ (pronounced “mew”), is calculated by using all the values in the population. The population mean is a parameter.

$$\mu = \frac{X_1 + X_2 + X_3 + \cdots + X_N}{N} = \frac{\Sigma X}{N}$$

where N represents the total number of values in the population.

EXAMPLE 3-1 Police Incidents

The number of calls that a local police department responded to for a sample of 9 months is shown. Find the mean. (Data were obtained by the author.)

475, 447, 440, 761, 993, 1052, 783, 671, 621

SOLUTION

$$\begin{aligned}\bar{X} &= \frac{\sum x}{n} = \frac{475 + 447 + 440 + 761 + 993 + 1052 + 783 + 671 + 621}{9} \\ &= \frac{6243}{9} \approx 693.7\end{aligned}$$

Hence, the mean number of incidents per month to which the police responded is 693.7.

EXAMPLE 3-2 Hospital Infections

The data show the number of patients in a sample of six hospitals who acquired an infection while hospitalized. Find the mean.

110 76 29 38 105 31

Source: Pennsylvania Health Care Cost Containment Council.

SOLUTION

$$\bar{X} = \frac{\sum X}{n} = \frac{110 + 76 + 29 + 38 + 105 + 31}{6} = \frac{389}{6} = 64.8$$

The mean of the number of hospital infections for the six hospitals is 64.8.

EXAMPLE 3–3 Miles Run per Week

Using the following frequency distribution (taken from Example 2–7), find the mean. The data represent the number of miles run during one week for a sample of 20 runners.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	Total 20

The completed table is shown here.

A Class	B Frequency f	C Midpoint X_m	D $f \cdot X_m$
5.5–10.5	1	8	8
10.5–15.5	2	13	26
15.5–20.5	3	18	54
20.5–25.5	5	23	115
25.5–30.5	4	28	112
30.5–35.5	3	33	99
35.5–40.5	2	38	76
	$n = 20$		$\Sigma f \cdot X_m = 490$

Find the sum of column D.

Divide the sum by n to get the mean.

$$\bar{X} = \frac{\Sigma f \cdot X_m}{n} = \frac{490}{20} = 24.5 \text{ miles}$$

Act

The Median

The **median** is the midpoint of the data array. The symbol for the median is MD.

Procedure Table

Finding the Median

- | | |
|---------------|--|
| Step 1 | Arrange the data values in ascending order. |
| Step 2 | Determine the number of values in the data set. |
| Step 3 | <i>a.</i> If n is odd, select the middle data value as the median.
<i>b.</i> If n is even, find the mean of the two middle values. That is, add them and divide the sum by 2. |

Range

The range is the simplest of the three measures and is defined now.

The **range** is the highest value minus the lowest value. The symbol R is used for the range.

$$R = \text{highest value} - \text{lowest value}$$

EXAMPLE 3-16 Comparison of Outdoor Paint

Find the ranges for the paints in Example 3-15.

SOLUTION

For brand A, the range is

$$R = 60 - 10 = 50 \text{ months}$$

For brand B, the range is

$$R = 45 - 25 = 20 \text{ months}$$

Make sure the range is given as a single number.

The range for brand A shows that 50 months separate the largest data value from the smallest data value. For brand B, 20 months separate the largest data value from the smallest data value, which is less than one-half of brand A's range.

The **population variance** is the average of the squares of the distance each value is from the mean. The symbol for the population variance is σ^2 (σ is the Greek lower-case letter sigma).

The formula for the population variance is

$$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N}$$

where X = individual value

μ = population mean

N = population size

The **population standard deviation** is the square root of the variance. The symbol for the population standard deviation is σ .

The corresponding formula for the population standard deviation is

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\Sigma(X - \mu)^2}{N}}$$

EXAMPLE 3-18 Comparison of Outdoor Paint

Find the variance and standard deviation for the data set for brand A paint in Example 3-15. The number of months brand A lasted before fading was

10, 60, 50, 30, 40, 20

SOLUTION

Step 1 Find the mean for the data.

$$\mu = \frac{\sum X}{N} = \frac{10 + 60 + 50 + 30 + 40 + 20}{6} = \frac{210}{6} = 35$$

Step 2 Subtract the mean from each data value ($X - \mu$).

$$\begin{array}{lll} 10 - 35 = -25 & 50 - 35 = +15 & 40 - 35 = +5 \\ 60 - 35 = +25 & 30 - 35 = -5 & 20 - 35 = -15 \end{array}$$

Step 3 Square each result $(\bar{X} - \mu)^2$.

$$\begin{array}{lll} (-25)^2 = 625 & (+15)^2 = 225 & (+5)^2 = 25 \\ (+25)^2 = 625 & (-5)^2 = 25 & (-15)^2 = 225 \end{array}$$

Step 4 Find the sum of the squares $\sum(\bar{X} - \mu)^2$.

$$625 + 625 + 225 + 25 + 25 + 225 = 1750$$

Step 5 Divide the sum by N to get the variance $\frac{[\sum(\bar{X} - \mu)^2]}{N}$.

$$\text{Variance} = 1750 \div 6 \approx 291.7$$

Step 6 Take the square root of the variance to get the standard deviation. Hence, the standard deviation equals $\sqrt{291.7}$, or 17.1. It is helpful to make a table.

Column A contains the raw data X . Column B contains the differences $X - \mu$ obtained in step 2. Column C contains the squares of the differences obtained in step 3.

A Values X	B $X - \mu$	C $(X - \mu)^2$
10	-25	625
60	+25	625
50	+15	225
30	-5	25
40	+5	25
20	-15	225
		1750

EXAMPLE 3-19 Comparison of Outdoor Paint

Find the variance and standard deviation for brand B paint data in Example 3-15. The months brand B lasted before fading were

35, 45, 30, 35, 40, 25

SOLUTION

Step 1 Find the mean.

$$\mu = \frac{\Sigma X}{N} = \frac{35 + 45 + 30 + 35 + 40 + 25}{6} = \frac{210}{6} = 35$$

Step 2 Subtract the mean from each value, and place the result in column B of the table.

$$\begin{array}{lll} 35 - 35 = 0 & 45 - 35 = 10 & 30 - 35 = -5 \\ 35 - 35 = 0 & 40 - 35 = 5 & 25 - 35 = -10 \end{array}$$

Step 3 Square each result and place the squares in column C of the table.

Step 4 Find the sum of the squares in column C.

$$\Sigma(X - \mu)^2 = 0 + 100 + 25 + 0 + 25 + 100 = 250$$

Step 5 Divide the sum by N to get the variance.

$$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N} = \frac{250}{6} = 41.7$$

Step 6 Take the square root to get the standard deviation.

$$\sigma = \sqrt{\frac{\Sigma(X - \mu)^2}{N}} = \sqrt{41.7} \approx 6.5$$

Hence, the standard deviation is 6.5.

A X	B $X - \mu$	C $(X - \mu)^2$
35	0	0
45	10	100
30	-5	25
35	0	0
40	5	25
25	-10	100

Formula for Sample Variance & Standard Deviation

Formula for the Sample Variance

The formula for the sample variance (denoted by s^2) is

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{n - 1}$$

where X = individual value

$\bar{X}$ = sample mean

n = sample size

Formula for the Sample Standard Deviation

The formula for the sample standard deviation, denoted by s , is

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

where X = individual value

$\bar{X}$ = sample mean

n = sample size



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Thank You

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